Multi-Agent LLM Evaluation of Counterfactual Explanations for Income Prediction

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Abstract

This project studies [selected AI method] in the context of [seminar topic]. We formulate the research question: [insert research question]. We review the relevant literature, implement the method, and evaluate it on [dataset/task]. Our results show that [main result], while also highlighting [main limitation].

Introduction and Problem Formulation

This project connects two seminar themes: **Counterfactuals for Explainability and Fairness in Machine Learning** and **Orchestrate, Secure & Scale Enterprise GenAI Architecture**. It is related to the first seminar because we study counterfactual explanations for a binary income prediction model. Given a person whose income is predicted to be below $50K, we use DiCE to generate counterfactual examples that would change the model prediction to above $50K. The central question is not only whether these counterfactuals flip the model output, but whether they are good explanations. In the context of explainable AI, a useful counterfactual should be close to the original input, plausible, actionable, and sparse, meaning that it should change only a small number of features.

The project is also related to **Orchestrate, Secure & Scale Enterprise GenAI Architecture** because we implement a multi-agent LLM pipeline for evaluating these explanations. The seminar discussed generative AI workflows, orchestration of AI services, hallucination control, auditability, human validation, and trustworthy LLM systems. Our pipeline follows this perspective by using several LLM-based roles, including a Prosecutor, Defense agent, Expert Witness, Judge, and Single Evaluator baseline, to assess counterfactual quality in a structured and auditable way.

The main research question addressed in this report is:

> *Do LLM-based evaluators provide added value over deterministic metrics for assessing counterfactual explanation quality, and does adversarial multi-agent debate improve over a single-LLM evaluator?*

This question is relevant because counterfactual generation methods such as DiCE typically select counterfactuals using deterministic criteria. These include **validity**, which checks whether the generated counterfactual flips the model prediction; **proximity**, which favors small changes from the original input; **sparsity**, which favors changing few features; and **feasibility**, which can restrict which features are allowed to change. These metrics are necessary, but they do not fully capture whether a proposed change is realistic, fair, causally coherent, or convincing from a human perspective. We therefore test whether LLM-based evaluators provide additional useful assessment of the counterfactuals generated by DiCE. Since no external ground-truth labels exist for the quality of these counterfactual explanations, we use a deterministic metrics-only evaluator as a rule-based reference standard. This reference encodes our project-specific human judgment through explicit heuristics and thresholds over validity, sparsity, proximity, confidence, and plausibility checks. We then compare a single-LLM evaluator and a multi-agent adversarial debate system against this reference.

We evaluate this question on the OpenML Adult Income dataset, where the task is to predict whether a person's income is above or below $50K.

Related Work

Taxonomy of the Research Area

Machine learning is increasingly used in high-impact domains such as healthcare, finance, and banking. This makes explainability important, because many useful models are difficult to interpret directly. The research area can be divided into three relevant families of methods:

- **Intrinsically interpretable models:** methods such as linear models or decision trees are easier to inspect because their decision process is relatively transparent. However, they may be less flexible than more complex models.
- **Post-hoc counterfactual explanations:** these methods explain a model decision by asking what minimal changes to the input would have led to a different prediction. They are useful because they can explain individual decisions without requiring access to the internal

structure of the model.

- **Counterfactual quality evaluation:** generated counterfactuals are usually assessed with metrics such as validity, proximity, sparsity, diversity, and feasibility. These metrics are useful but incomplete, because they do not always capture whether a recommendation is realistic, fair, or actionable in the real world.

This taxonomy positions our project in the third family. We do not only generate counterfactual explanations; we study how their quality should be evaluated. In particular, we compare deterministic quality metrics with an LLM-based multi-agent audit that reasons about the same counterfactuals from several perspectives.

Seminal and Recent Works

Counterfactual explanations were introduced as a way to explain automated decisions without opening the black box of the underlying model (Wachter, Mittelstadt, and Russell 2017). Instead of describing all internal model mechanisms, a counterfactual explanation answers a local question: what would need to change in this input for the model to produce a different outcome? For a classification task, this means showing how an unfavorable prediction could be changed into a favorable one.

Later work extended this idea by emphasizing diversity and actionability, rather than validity alone. DiCE generates multiple diverse counterfactual alternatives for a given instance, giving users several possible paths toward the desired outcome instead of a single potentially unrealistic modification (Mothilal, Sharma, and Tan 2020). This is the counterfactual generation framework used in our project.

However, DiCE-style evaluation still relies heavily on numerical criteria. Validity checks whether the model prediction flips; proximity measures how close the counterfactual is to the original instance; sparsity measures how many features change; diversity measures variation among generated alternatives; and feasibility constrains which features may be modified. These criteria are necessary, but they can miss important semantic failures. For example, a counterfactual may be close in feature space while still proposing an impossible change, such as decreasing a person's age from 30 to 25. Similarly, numerical metrics may not fully capture fairness, realism, or causal coherence.

Our project addresses this gap with a multi-agent collaborative audit of counterfactual explanations. Because there is no external ground truth for these quality labels, our deterministic metrics-only evaluator acts as the reference standard: it encodes our human judgment through explicit rules over metrics and issue-specific plausibility checks. The LLM evaluators are then assessed by how well they reproduce, justify, or meaningfully challenge this rule-based assessment. To make the audit operational, we define a small issue taxonomy covering implausible time-dependent changes, extreme working hours, unactionable capital shifts, too many simultaneous changes, and fragile counterfactuals; the next section describes how these labels are used in the evaluation pipeline.

Method

We now describe [selected method]. The method aims to [goal of method]. At a high level, it works by [main algorithmic idea].

Given an input x, the method computes [output] by solving/estimating:

$$\hat{y} = f_\theta(x), \tag{1}$$

where f_θ denotes [model/method]. The key steps are:

Step 1 ;
Step 2 ;
Step 3 .

In our implementation, we follow [paper/source] with the following adaptations: [describe simplifications, modifications, or implementation choices].

Experimental Setup

Dataset

We evaluate the method on [dataset name]. The dataset contains [brief description: number of samples, features, classes, task, etc.].

Baselines

We compare [selected method] against the following baselines:

- **Baseline 1:** [description];
- **Baseline 2:** [description].

Metrics

We use the following metrics:

- **Metric 1:** [why appropriate];
- **Metric 2:** [why appropriate];
- **Metric 3:** [why appropriate].

Implementation Details

The implementation is written in Python. The code is available at:

https://github.com/your-username/your-repository

The repository includes instructions to reproduce the experiments, including dependencies, dataset preparation, and commands used to generate the results.

Results and Analysis

Table 1 summarizes the main experimental results.

The results show that [main observation]. Compared with [baseline], the selected method [performs better/worse] in terms of [metric]. This suggests that [analysis].

However, the method also presents limitations. First, [limitation 1]. Second, [limitation 2]. These limitations are important because they affect [robustness, scalability, interpretability, feasibility, etc.].

Figure **??** illustrates [what the figure shows].

Method	Metric 1	Metric 2	Metric 3
Baseline 1	–	–	–
Baseline 2	–	–	–
Selected method	–	–	–

Table 1: Experimental results comparing the selected method with baselines.

Discussion

The experiments indicate that [answer to research question]. The main strength of [selected method] is [strength]. This makes it suitable for [setting].

Nevertheless, the method assumes that [assumption]. In practice, this assumption may not hold when [case]. Another limitation is [another limitation].

Possible improvements include [future improvement], [extension], or combining the method with [other approach].

Conclusion

This report studied [selected method] for [task/problem]. We reviewed the relevant literature, implemented the method, and evaluated it experimentally. Our findings suggest that [main conclusion]. Future work could investigate [future direction].

References

Mothilal, R. K.; Sharma, A.; and Tan, C. 2020. Explaining Machine Learning Classifiers through Diverse Counterfactual Explanations. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 607–617. ACM.

Wachter, S.; Mittelstadt, B.; and Russell, C. 2017. Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GDPR. *Harvard Journal of Law & Technology*, 31(2): 841–887.